

Saiansh Chaddha

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EDUCATION

Stony Brook University

Master of Science in Computer Science

- Concentration in Data Science and Engineering

Bachelor of Science in Computer Science, Applied Mathematics & Statistics

- Dean's List, Magna Cum Laude Distinction

Stony Brook, NY

Aug. 2023 – May 2025

Aug. 2019 – May 2023

EXPERIENCE

Software Engineering Teaching Assistant

Stony Brook University

- Led as head teaching assistant for 115 students, ensuring organized exam proctoring and successful course delivery
- Supported student success through 3+ weekly office hours, active class engagement, and 100+ Piazza forum responses, increasing accessibility and comprehension across course concepts
- Enhanced collaborative learning by developing 7 Process Oriented Guided Inquiry Learning (POGIL) worksheets

Jan. 2023 – May 2023

Stony Brook, NY

Software Engineer Research Assistant

Stony Brook University

- Developed a full-stack interactive display for 200+ historical postcards under the supervision of Dr. Joydeep Mitra
- Created a user-friendly Leaflet-based map to visualize the 200+ postcards from 25 countries, enhancing historical context with 8 color-coded empire categorizations
- Applied Agile methodology and Git for version control in weekly researcher collaborations via Slack/Zoom

Aug. 2022 – Jan. 2023

Stony Brook, NY

Software Engineer Intern

Data Device Corporation

- Developed and delivered a comprehensive 25+ page GUI software verification document using black box testing, ensuring robust deployment readiness and functional validation
- Debugged legacy Visual Basic code, successfully verifying 50+ International Space Station chips, ensuring compliance with space-grade standards
- Optimized robotic system precision, attaining 98% placement accuracy by training robot on 800 company IC chips
- Accelerated CSV parsing by 70% for streamlined data processing by implementing Python Regex pattern matching

June 2021 – Jan. 2022

Bohemia, NY

PROJECTS

Distributed Banking System with Paxos Protocol | *Java, Multithreading, Paxos, Git* Sept. 2024 – Oct. 2024

- Processed 500+ transactions across 5 servers, ensuring fault tolerance and reliability through advanced multithreading in Java
- Ensured 100% consensus accuracy in resolving transaction conflicts, implementing the Paxos protocol for reliable data consistency in the distributed system
- Decreased transaction latency by 50% using thread-safe multithreading, optimizing application responsiveness

AgentBert: Chatbot with Toxicity Detection | *Python, PyTorch, HuggingFace* March 2024 – May 2024

- Engineered a toxicity-filtering chatbot achieving 93% accuracy, leveraging GPT-2 and custom toxicity detection models, trained on a dataset of 100,000 toxic/non-toxic conversations
- Implemented a toxicity detection model by combining GPT-2 and HateBERT using a weighted average approach
- Developed an alternative chatbot model trained with T5 on toxic datasets, classifying input/output for enhanced toxic conversation handling

A Digital Platform for Showcasing Historical Postcards | *React.js, Node.js, CSS, Git* Aug. 2022 – Jan. 2023

- Designed a MERN-based interactive platform for displaying 200+ historical postcards, advancing user engagement and accessibility of historical data
- Presented the interactive platform to 90+ users, receiving positive feedback on the Leaflet-based world map feature
- Interactive platform available at <https://empirehistorycards.cs.stonybrook.edu/>

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, HTML/CSS, Ruby, Go, Bash, C++

Frameworks: React.js, Node.js, D3.js, Express.js, Material-UI, MongoDB, SQL

Developer Tools: Git, GitHub, AWS, GCP, VSCode, PyCharm, Google Colab, IntelliJ, Eclipse, Jupyter

Libraries: PyTorch, Pandas, Scikit-Learn, Matplotlib, Pytest, HuggingFace, Axios, Mongoose

Certifications: J.P. Morgan - Software Engineering Job Simulation